

Contribution: Documentation of modern IoT botnet behavior and mitigation strategies [5].

- Investigate the underlying code architecture to determine major functions and operation.
- Identify and assess DDoS attack techniques and payload delivery methods [3].
- Evaluate multi-architecture and cross-platform support mechanisms.
- Recommend mitigation methods for discovered vulnerabilities [4].

Vulnerabilities Found: Buffer overflows (`sprintf`), no thread sync, hardcoded credentials.

The architecture follows a centralized C2 model [1]. The `server.c` component manages the botnet via TCP port 1994, while `bot.c` resides on infected devices. The `cc.py` build system is crucial for cross-compiling payloads for various architectures.

The infection cycle begins with network scanning (`scankill.c`) to identify vulnerable devices. Upon successful exploitation (e.g., via Telnet brute-force or specific CVEs), the payload is downloaded and executed, establishing a persistent connection to the C2 server [2].

High Risk: Hardcoded IP addresses allow defenders to sinkhole the entire botnet. **Medium Risk:** Lack of encryption enables traffic inspection and signature generation.

Unlike early IoT botnets, SlumX supports 12 distinct architectures, significantly increasing the potential infection pool [3][5].

The botnet supports multiple DDoS vectors. Analysis shows that HTTP floods are the most complex to implement (95 LOC), requiring valid header construction to bypass simple filters. In contrast, UDP and TCP floods rely on raw socket manipulation for high-volume traffic generation [3].

- 1 M. Antonakakis et al., "Understanding the Mirai Botnet," *USENIX Security*, 2017.
- 2 A. Marzano et al., "The evolution of bashlite and mirai IoT botnets," *IEEE S&P*, 2018.
- 3 G. Vishwakarma and G. S. Chhabra, "Targeted DDoS Attacks on IoT: A Survey of the Mirai Botnet and its Variants," *AISC*, 2023.
- 4 A. D. Alghazzawi et al., "An Efficient Hybrid Deep Learning Model for Detecting IoT Botnet Attacks," *IEEE Access*, 2024.
- 5 Statista, "IoT connected devices installed base worldwide," 2024.

